P. Project
PROPOSAL

Submitted to:

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Project Proposal

Algorithmic Trading Strategy Back-tester with Chatbot

Project Title	**Algorithmic Trading Strategy Back-tester with Chatbot**
Course	Programming for Business (C++)
Semester	Spring 2026
Instructor	Muhammad Tehaam
Submission Date	28th February 2026

Team Members

Member	**Name**	**Student ID**	**Role**
Member A	Sarah Shabbir	25i-5554	Data Loader & Structures
Member B	Syed Ayan Ali	25i-5639	Indicators & Signal Generation
Member C	Syed Hamza	25i-5558	Backtest Engine & Metrics
Member D	Awais Qasim	25i-5647	Reporting & Chatbot

1. What Are We Building?

We are building a **C++ program that tests stock/crypto trading strategies** using historical data:

- You start with **$10,000 fake (only numbers) money**.
- The program looks at **past stock/crypto prices** (from a CSV file).
- It follows a simple rule to decide when to **buy** and when to **sell**.
- At the end, it shows you **how much money you made or lost**.
- You can also **ask questions** about the results, like "How many losing trades did I have?" and the program will answer.

This is called a **backtester.** We test a trading strategy on old data to see if it worked.

2. Why Are We Building It?

- To practice **C++ programming** using what we will learn in course (arrays, functions, pointers, file reading, etc.).
- To understand how **trading strategies** work in the real world.
- To create something **useful and interesting** that we can show in our portfolio.
- To work as a **team** and learn how to divide a project into smaller pieces.

3. What Will the Program Do?

Here is what the user will experience:

1. **Run the program** – it loads stock/crypto data from a file.
2. **Backtest runs automatically** – the program simulates trading for the entire period.
3. **See results** – a summary shows total profit, number of trades, win rate, etc.
4. **Chat mode starts** – the user can type questions and get answers.
5. **Type 'exit' to quit** – program ends.

4. How Will We Build It? (The Simple Version)

We will split the work into **4 modules**, one for each team member. Each part is like a puzzle piece that fits together.

Module 1: Data Loader – *Sarah Shabbir*

Job: Read the stock/crypto data from a CSV file and store it neatly.

- Open a CSV file (like Excel) that contains dates and prices.
- Read each line and save the information OHLCV (date, open price, high price, low price, close price, volume) into an array.
- Make sure the data is correct (no missing values, dates in order).
- Provide helper functions so other members can easily get a price for any day.

Module 2: Indicators & Signals – Syed Ayan Ali

Job: Look at the prices and decide when to buy and sell.

- Calculate **moving averages (Simple / Exponential)** – the average price over the last 20 days and last 50 days.
- Apply the **trading rule**:
  - When the 20-day average goes **above** the 50-day average → **BUY**.
  - When the 20-day average goes **below** the 50-day average → **SELL**.
  - Otherwise → **HOLD**.
- Store these decisions (buy, sell, hold) in an array for each day.

NOTE: We can make it dynamic by adding more strategies like RSI, MACD, etc.

Module 3: Backtest Engine – Syed Hamza

Job: Simulate the trading and calculate how much money we make.

- Start with $10,000 cash and 0 shares.
- Go day by day:
  - If the signal says BUY and we have no shares → buy as many as we can with our cash.

  - If the signal says SELL and we have shares → sell all of them.
  - Keep track of cash, shares, and portfolio value each day.
- Record every completed trade (entry date, exit date, buy price, sell price, profit).
- At the end, calculate performance numbers:
  - **Total return** (how much money we made in $ / %)
  - **Sharpe ratio** (a measure of risk – higher is better)
  - **Maximum drawdown** (biggest drop from a peak)
  - **Win rate** (% of trades that made money)
  - And more...

Module 4: Reporting & Chatbot – Awais Qasim

Job: Show the results nicely and let the user ask questions.

- Print a **summary table** with all the important numbers (like a report card).
- Use lines and boxes (╔═╗) to make it look professional.
- Create a **chat loop** that asks the user to type and waits for their question.
- Recognize keywords like "total return", "losing trades", "help", etc.
- For each question, print the answer by looking at the data from Module 3.
- If the question is not recognized, say "Sorry, I don't understand. Type 'help'."

NOTE: But if possible, we make ***LLM model*** to answer questions.

5. How the Modules Fit Together

Module A (Data Loader)

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Module B (Signals): needs price data from Sarah

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Module C (Backtest): needs price data from Sarah and signals from Ayan

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Module D (Chatbot): needs trade data and metrics from Hamza

We will work **in parallel** by using temporary fake data until the real modules are ready. Then we connect them one by one.

6. What We Will Learn

- How to **work in a team** on a real project.
- How to **split a problem** into smaller pieces.
- How to **read from files** and process data.
- How to **write functions** that do specific jobs.
- How to **use arrays** to store lots of information.
- How to **make a simple chatbot** that understands keywords.
- How to **present results** in a clean, professional way.

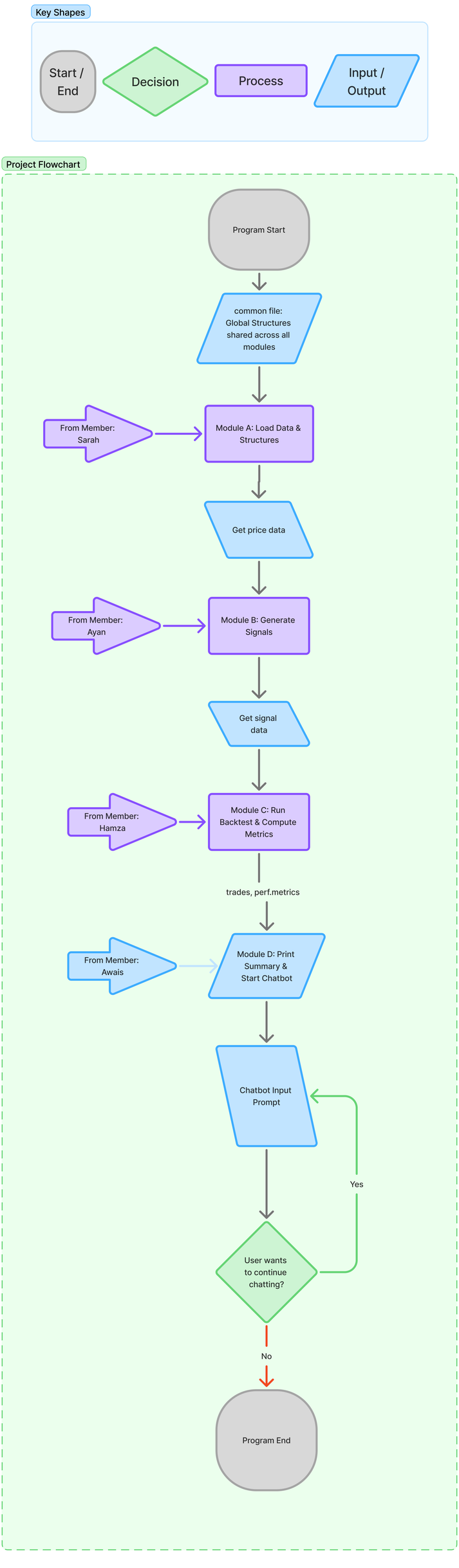
7. Final Words

This project is challenging but **doable**. Each person has a clear job, and we will help each other when someone gets stuck. By the end, we will Insha'Allah have a working program.

Let's build something awesome with the help of members!

Simple Glossary (if you get confused)

Term	**Simple Meaning**
Backtest	Testing a trading strategy on old data to see if it works before risking real money.
Moving Average	Average price over a certain number of days (smooths out ups and downs).
Signal	A decision: BUY, SELL, or HOLD.
Sharpe Ratio	A number that tells you if profits are worth the risk. Higher is better.
Drawdown	How much your money dropped from its highest point.
Win Rate	Percentage of trades that made money.
CSV File	A simple text file with values separated by commas (like a mini Excel sheet).


Key Shapes
Start / End
Decision
Process
Input / Output
Project Flowchart
Program Start
common file: Global Structures shared across all modules
From Member: Sarah
Module A: Load Data & Structures
Get price data
From Member: Ayan
Module B: Generate Signals
Get signal data
From Member: Hamza
Module C: Run Backtest & Compute Metrics
trades, perf.metrics
From Member: Awais
Module D: Print Summary & Start Chatbot
Chatbot Input Prompt
Yes
User wants to continue chatting?
No
Program End